Function Spaces

B. Math (Hons.) III Year

End-Semester Exam, November 2025

Maximum Score: 50 Duration: 3 hours

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1. Determine whether the following statements are true or false (give justifications).

   (i) There is a Riemann integrable function f such that

$$\hat{f}(n) = \begin{cases} \frac{1}{\sqrt[3]{n}}\,; & n \in \mathbb{N} \\ 0\,; & \text{otherwise} \end{cases}$$

   (ii) Uniform limit of a sequence of contractions $\{T_n\}$ on a set $X \subseteq \mathbb{R}$ is always a contraction.

   (iii) Every continuous function $f \in C[a,b]$, where $a, b \in \mathbb{R}$ and $0 \notin [a,b]$, can be uniformly approximated by sequence of odd polynomials. [Marks: $3+3+4$]

2. Let $f : [0,1] \to \mathbb{R}$ be Riemann integrable and consider the equation

$$f(x) = x + \lambda \int_0^1 (x^2 + y^2) f(y)\, dy \qquad (\lambda \in \mathbb{R}).$$

   Show that, the above equation has a unique solution in $C[0,1]$ if $|\lambda| < \frac{1}{2}$. [Marks: 4]

3. Let $f_n : [0,1] \to \mathbb{R}$ be continuous functions such that

$$|f_n'(x)| \leq \frac{1}{\sqrt{x}} \quad \text{and} \quad \int_0^1 f_n(x)\, dx = 0$$

   for every $n \in \mathbb{N}$. Prove that $\{f_n\}$ has a uniformly convergent subsequence. [Marks: 5]

4. Let $(X, \mathcal{M}, \mu)$ be a measure space, $\{E_n\} \subseteq \mathcal{M}$ and

$$E := \{x \in X : x \in E_n, \text{ for infinitely many } n\}.$$

   Prove that,

   (i) $E \in \mathcal{M}$.

   (ii) If $\sum_{n=1}^{\infty} \mu(E_n) < \infty$ then $\mu(E) = 0$. [Marks: $3+3$]

5. State and prove Fatou's lemma. [Marks: 5]

6. Let $(\mathbb{R}, \mathcal{L}, m)$ be the Lebesgue measure space and $E \in \mathcal{L}$ such that $m(E) > 0$. Prove that, $\exists\, F \subseteq E$ such that $F \notin \mathcal{L}$. [Marks: 5]

7. Consider the following function on the circle:

$$f(x) = \begin{cases} 0\,; & -\pi \leq x \leq 0 \\ \sin x\,; & 0 < x \leq \pi. \end{cases}$$

   (i) Find the Fourier series of f.

(ii) Using the Fourier series of f, deduce that

$$\frac{1}{1\times 3} - \frac{1}{3\times 5} + \frac{1}{5\times 7} - \cdots = \frac{\pi - 2}{4}.$$

[Marks: $4+3$]

8. Considering the function $f(x) = x$ on $[-\pi, \pi)$ and using the Parseval's identity deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

[Marks: 4]

9. Let f and g be two Riemann integrable functions on the circle. Prove that $f * g = g * f$ (where '$*$' denotes the convolution operation). [Marks: 6]